# The Development of History Educational Game as a Revision Tool for Malaysia School Education

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**Abstract.** Computer game has indisputably become a culture especially among teenagers even in Malaysian community. Through the use of broadband and civilizing of IT usage, teenagers have now been exposed to computer game since at young age. A decade ago, computer game was perceived as only a medium for students to seek pleasure without giving any benefit to them. Recently, various attempts have been arranged to enable computer game to be incorporated into classroom to make learning experience more interesting and enjoyable. As a result, an educational computer game prototype has been developed for the purpose of meeting these requirements. The prototype, which is dedicated for the history subject for Malaysia secondary school, mainly serves as a revision tool for the student. MUDPY model has been employed as the main methodology for the project. The prototype was tested using the black box test to ensure the functionality and PUEU test for the effectiveness. This paper exposes the techniques used for development and the findings resulted from the tests conducted.

**Keywords:** educational games, history educational game, game prototype, and revision tools, Visual Informatics.

## 1 Introduction

The most crucial task in learning history subject is to overcome the boring phenomenon among students. Curriculum Development Division reported that history subject is known as a ‘dead’ and boring subject [2]. Normally, the society assumed that history subject lacks the commercial value. Students also don’t show much interest in learning the subject because they assume the history subject is dull and they have to memorize and understand all the facts, concept, time and historical events [14]. Nevertheless, history subject in Malaysia is a core subject in the Kurikulum Bersepadu Sekolah Menengah (KBSM) and must be well educated by all students on an ongoing basis for five years. Thus, the history curriculum was arranged to have continuity from Junior Secondary Schools (SMR) until Senior Secondary Schools (SMA) so that the foundations of knowledge, values, skills and experience gained can be strengthened and developed further [4].

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70.6 % students admitted that the main problem in learning history is the difficulty to memorize historical fact [11]. At the same time, 44.5% student has no interest in history because of teaching media such as dreary textbooks. Another 15.8 % student claimed that they are discouraged due to teachers' unexciting teaching method. Meanwhile lack of teaching aid/material used by the teacher contributed 16.5% of student negative perception in history learning. Student did not get clear description about historical events and cannot understand history context. Deficiency of history references also contributes to the existing problem.

In order to overcome the boring phenomenon, we developed an educational game prototype to integrate fun in history learning. The prototype can be used as teaching aid in class or as a revision tool at home. With the right development and use, an educational game could be a great tool in education [13]. Another study found that game has attractive elements, which include motivation and engagement [3]. Computer game can be exciting, fast paced, appealing, compelling and rewarding compared to classroom-based strategies and it provides an ideal learning environment for students [17].

The educational game prototype represents the history about Japanese invasion on Malaya as in the form 3-history syllabus of SMA. Realizing the positive impacts that educational game may has towards learning history in school, this paper will highlight the development phase in building the educational games prototype to enhance the learning process.

## 2 Related Research on Educational Game

Game is based on the concept of fun [13] and is a free activity, outside ordinary life, with no profit [18]. It has rules and a defined way of progressing. It may possess social groupings that cloak themselves in secrecy to stress their difference from the common world. Game may create a make-believe element [5] and it is an activity that is executed only for pleasure and without conscious purpose [10].

An educational game, one designed for learning, is a subset of both play and fun. It is a melding of educational content, learning principles, and computer games [13]. An educational game is combining the element of joy and learns in a playable situation. Serious game that is synonym with educational game can be defined as a mental contest, played with computer in accordance with specific rules that use entertainment to further government or corporate training, education, health, public policy and strategic communication objectives [18]. It is called serious game because it is used in a pedagogical way for political, social, marketing, economical, environmental or humanitarian purposes [1]. Serious game has not just a story, art, and software, but also involves pedagogy: activities that educate or instruct, thereby imparting knowledge or skill [18].

Although there are opinions saying game leads kids into violent, but the effectiveness of educational game can't be denied [9]. Many researchers have come out with results that show the positive sides of gaming in education. Game may provide limited superficial information and is not enough to satisfy young people's educational needs, but it may be enough for them to get a grasp on it and that in more overtly educational settings the role of teachers, peers and other supporting materials will be necessary to build on these superficial understandings [7]. Game is effective; not because of what it is, but because of what it embodies and what learners are doing as they play a game. Table 1 shows a portion of recent works by selected researchers . Based on the given evidence, we conclude that game can be effective for educational purposes and learning process becomes more interactive and enjoyable.

**Table 1.** An Overview of the Studies Related with Effectiveness in using Computer Games in Education [6][7]

| Author(s) | Year | Genre | N | Subject | Results |
|---|---|---|---|---|---|
| Noble et al. | 2000 | Action | 101 | Drug education | Students taught through video games find the experience motivating and want to play the video game again. |
| Din Feng & Caleo | 2000 | – | 47 | Spelling and math | Children who play video games learn (mostly in spelling) better compared to peers who do not use video games. |
| Turnin et al. | 2000 | – | 2000 | Eating habits | Video games can teach students about eating habits and lead to significant change in everyday habits. |
| Lieberman | 2001 | Action | | Asthma, diabetes, | A review of a number of research projects supports the notion of learning from video games. |
| Becker | 2001 | Action | – | Programing | The study testifies to the increased motivation in connection with video games. Games are found to be more effective and motivating than traditional teaching. |
| McFarlane et al. | 2002 | – | – | All subjects | The study finds that teachers in general are skeptical towards the learning of content with video games. However, teachers appreciate the learning of general skills. |
| Gander | 2002 | Strategy | 29 | Programing | The study finds that video games are especially effective for teaching specific knowledge. |
| Rosas et al. | 2003 | Action | 1274 | Reading and math | Video games increase motivation, and there is a transfer of competence in technology from using the video game. |
| Squire et al. | 2004 | Simulation | 96 | Physics | Students using the simulation game performed better compared to the control group. |
| Egenfeldt-Nielsen | 2005 | Strategy | 72 | History | Students initially learn the same in history when using video games but have better retention. |
| Buch & Egenfeldt-Nielsen | 2006 | RPG | 72 | Social Studies | 60% students on self-assessment found they learned more with Global Conflicts: Palestine than a traditional course. Almost 40% that it was around the same. |

## 3 Educational Game Prototypes: History of Japanese Occupation in Malaya

Development is an important phase in any project lifecycle. This phase determine wheter we can answer the problems that have been addressed in the ealier section. This project development phases are based on the Multimedia Design and Planning Pyramid (MUDPY) model [15]. An educational game prototype involves iterative process during development. The development process also comprises ideas taken from a wide range of fields, including marketing, pedagogy, digital technology, aesthetics, graphics, intellectual property law, and management.

MUDPY model consist of three phases; which are planning, design and production. There were several stages that we adopted in order to develop the prototype and these stages are shown in Figure 1.

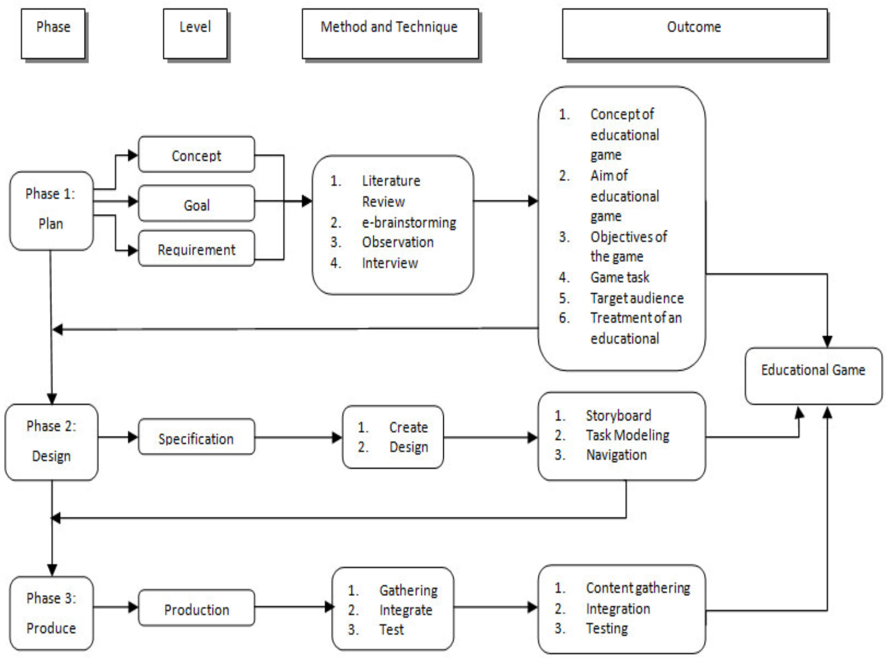


**Fig. 1.** The development cycle diagram

### 3.1 Phase 1: Plan

In this stage, concept, goal and requirement were gathered through literature reviews, e-brainstorming, observations and interviews. Based on literature review as explained in previous section, the concept of education game, previous reseacher works and education theories have been identified and integrated as the requirement for the development. Using e-brainstorming method, a set of questions related with educational game has been circulated through email to the teachers in order to gather some input for the development. Several history teachers were selected from all over the country for this reason. Observation was done in several school in Malaysia by sitting at the back of the classroom to experience the history learning process in the school. After the class end, we took several minutes to interview the teacher and selected students. The outcome was used to derive the concept, objectives, aim, task, target audience and game treatment.

**e-Brainstorming.** From the e-brainstorming, we found that the teachers had a problem with the student attention in the class. They confess that students are not interested in history because they had to memorize all the facts. Lack of teaching aids also contributed to the lack of students attention in the class. Teachers cannot find any good teaching aid to assist them in the class and they are willing to use any possible teaching aids if provided. They also state that multimedia teaching aids would be better compare to the other teaching aids because in the past multimedia was known to attract students attention.

Students perceptions on the teaching methods used had only worsen the current problem. They consider the methods to be outdated and traditional whereas they prefer modern teaching aids and technologies. With combination of better teaching methods and aids, a classroom can be a very conducive place to learn.

**Interview.** Based on the interview conducted in several schools, the findings are concluded into two parts as stated below.

*Teachers.* Teachers had good experience in history and teaching, but lack of students initiative to make early preparations was a major drawbacks to the teachers. Teachers also want students to make their own learning aids like mind maps, but students did not take serious action to the task. From the interview teacher also willing to use a new teaching aid like educational game in class to help the knowledge transfer between students and teachers.

*Students*. From the interview, we found that the most attractive part of learning in school is the way of knowledge being delivered in class. Students ready to learn in a conducive environment. They don't want a teacher that always gave them pressure to study and create unease feeling in the classroom. Students also like to learn something that have impact on their life. From the interview, we also found that students spends an average of 1.5 hours a day to play video game. Sports game is the most favorable game for them. Second most played game is a 3D first person shooter game.

**Observation.** Based on the observation that had been done, we have concluded that the history class in the school can be more fun by adding some interesting value into the teaching methods. Teachers always used chalk and blackboard as a major tools which is very unexciting. Students only write down the note without understanding the contents. Teacher also don't emphasis on the important facts.

**Requirement Derived from Phase 1.** In this phase, a set of requirement had been produced based on the information gathered through literature reviews, e-brainstorming, interviews and observation. The requirement list of the educational game prototype is shown in Table 2. This table includes the concept, aim, objectives, task, target audience and game treatment for the prototype development.

**Table 2.** Requirement for Educational Game Prototype

| No | Subject | Outcome |
|---|---|---|
| **1.** | Concept | This educational game is about the invasion of Japanese army in Malaya. Student have to answer question to eliminate enemy and win the game. Mark will be given based on the right answer. |
| **2.** | Aim | To make the history learning experience as fun as possible and deliver knowledge at the same time. |
| **3.** | Objectives | After playing the game student can:<br>1. State the reasons for Japan's expanding power.<br>2. Describe the arrival of the Japanese to Malaya.<br>3. List the success factors of Japan dominates Malaya. |
| **4.** | Task | Student can walk in the game room and shoot the enemy to answer question (multimedia). Student can learn about the facts related to Japan invasion (learning) |
| **5.** | Target audience | Form 3 student in Malaysia Secondary School |
| **6.** | Game treatment | Game type is first person perspective. Game must have smooth look and high frame per seconds. |

### 3.2 Phase 2: Design

This phase is also known as specification level [15]. In this phase, we designed the character, storyboard, task modelling and also navigations. Specifications for the prototype such as platform, hardware and software have also been identified. The chosen platform for this prototype development is a Windows base due to the student awareness in using Windows platform at home. Any computer using Intel Pentium 4 processor is suitable to run the prototype application. The memory should be 1MB or above and there should be a free space of 500 KB on the hard disk. The computer should be occupied with multimedia devices such as CD-ROM, speaker, monitor, keyboard and mouse.

**Character Design.** Character design was done through sketching process. Then, the sketches were transfered into a computer through a scanner. Manipulation of the characters was done through Adobe Photoshop. Characters were selected carefully to give an impact to the student while playing this game. The antagonist character was pictured as Japanese army, who invaded Malaya while the protogonist character was pictured as Malay army that secured the Malaya in that particular time.

**Storyboard.** After the character design, storyboard was created. This storyboard contains the storyline inside the game. The storyboard is a vital document in this game development. The storyboard will determine the flow and elements of the prototype and it becomes the guide for the researchers. The welcome screen was design with simple instructions to ensure students understanding and good performance while playing the game.

Figure 2 shows the screen when the player engages with an enemy. To enable the player to shoot the enemy, the player must answer a pop-up question as show in Figure 3. The player has to pick one correct answer. If the answer was correct then the player will get a bonus mark and if it was incorrect then the player will be given one more chance. If the answer was accurate on the second try, a normal score will be given but if the answer was inaccurate the cumulative score and health will be deducted. If the health value becomes zero after deductions, then the game will be over. The player has to encounter ten enemies and responds to ten questions for the game to be completed. Final score will be shown at the last stage when the game is completed or if the game is over.

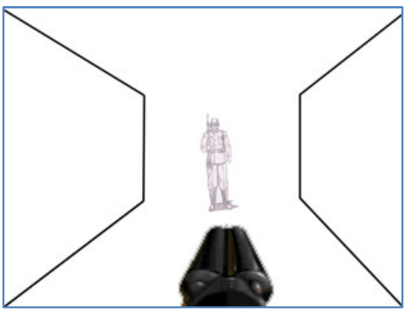

**Fig. 2.** Player engages with an enemy

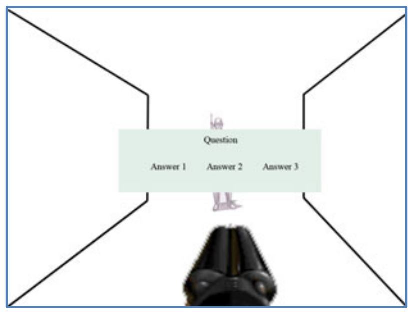


**Fig. 3.** A pop-up question

**Task modelling.** Task Modeling techniques provide the means to specify goal-oriented tasks [12]. It describes the window for required interfaces. Task modeling has some influence on the game composition and navigation. Figure 4 shows an example of a task flow for the game. The player must complete the current task in order to get to the next task. In this game prototype, the task is to answer the multiple-choice question. Player is given two chances to answer the question. After answering the question correctly, player can continue playing the game.

**Game Navigation.** Navigation outlines on how the pages in the game link together. There are several ways of navigation like linear, circular, network and tree structure. For this game prototype, linear navigation was used because there is no upper level. Player can end the game anytime by hitting the escape key. The structure of game navigation is shown in Figure 5.

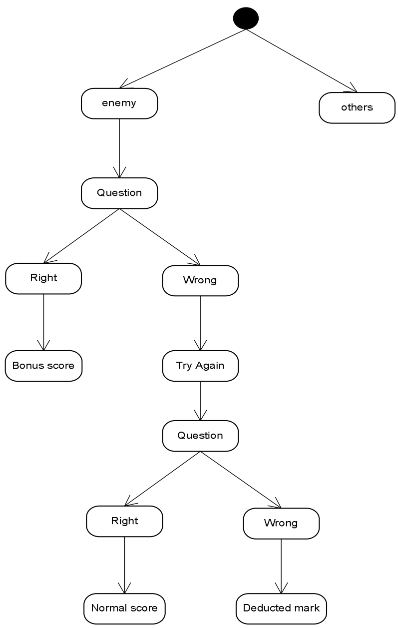


**Fig. 4.** Task Modeling

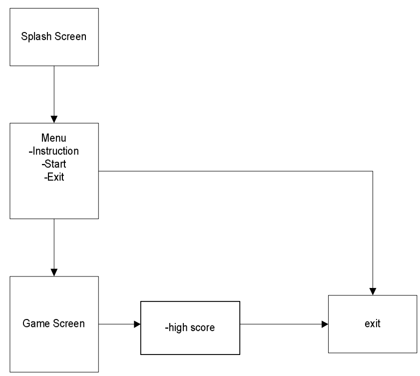


**Fig. 5.** Game Navigation

### 3.3 Phase 3: Produce

In this stage all the components are integrated together to produce a game to fullfill the objectives. The methods used are gathering, integrating and testing. The outcome of every phases are combined to produce the educational game prototype. Components like graphic, audio, animation, video and text are selected to ensure the product greatest functionality.

**Content Gathering.** The content of the game is gathered through various ways. Most of the content are gathered through the Internet and history textbook. Then the content is integrated together using Game Maker 8.0 software. Before the content can be implanted into the game maker software, some of them have to be built and edited.We used Adobe Photoshop editing software to edit pictures. After that, we integrated all items into the sprites and converted them as objects in the Game Maker 8.0 software. The content were blended properly in order to create useable prototype which meets the initial objectives.

**Integration.** Sprite is the main component inside Game Maker 8.0; where all the animations and the characters are placed. Sprite is created like an animation to make sure the movement of the character is as smooth as it can be. After that, the sprite was transformed into an object in which we can manipulate it's behavior and condition using some programming codes.

Proper background and wall were used to form the game border. This background and wall were the environments choosen to make sure the condition of the game was synchronized to the storyline. In this game, researchers had decided to used the jungle theme as the game was based on the war in the jungle area. Sound is added and integrated together into the Game Maker software. This sound is edited using Audacity Software and Sony Sound Forge.

In the Game Maker software there is a pane known as a room. Room is the place where all the resources is integrated together. Before the room can be used , we must drew a border through a wall and a background. The room is viewed from the top to make sure we know where was the placed to put the player, object, obstruction and enemies. Finally we were succeed in developing the game prototype. Two screen shots sample of the game interfaces can be viewed in Figure 6 and Figure 7.

**Fig. 6.** Screen shot of the educational game prototype

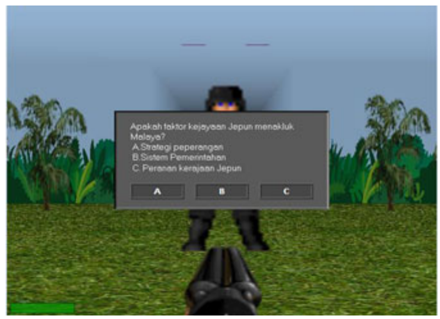


**Fig. 7.** Screen shot for the questions asked

**Testing.** Testing was done to ensure the prototype fulfilled the user needs and can perform the task properly. For this prototype, black box testing has been used for the functional system testing. Meanwhile for the usefulness of the system the Perceived Usefulness and Ease of Use (PUEU) test has been implemented. Black box testing has been piloted during the development for each phases. This test was rapidly conducted to make sure the prototype was working well and the development process was in the right track. After each test was completed, we made a few adjustments to meet the requirement of the prototype.

PUEU test was conducted at the end of development process, once the prototype was completely produced. A sample of twenty students was selected from various gender to participate in this survey. They were ask to fill up a questionnaire after playing the game prototype. Result of the test will determine whether the prototype had successfully satisfy the initial requirements. PUEU consists of twelve questions with seven scales from unlikely to likely.

To analyze the PUEU test, researchers carried out a descriptive analysis by using Microsoft Excel software. The questions were divided into two parts. The first part, question 1 to 6 related to the perceived of usefulness meanwhile the second part, question 7 to 12 related to the ease of use. Table 3 and 4 describe the results derived from the answered questionaires.

**Table 3.** Descriptive Analysis of Perceived Usefulness

| | Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Question 6 |
|---|---|---|---|---|---|---|
| **Mean** | 6.25 | 5.9 | 5.31 | 5.75 | 5.8 | 6.31 |
| **Standard Error** | 0.25 | 0.26 | 0.41 | 0.28 | 0.32 | 0.26 |
| **Median** | 7 | 6 | 6 | 6 | 6.5 | 7 |
| **Mode** | 7 | 7 | 6 | 7 | 7 | 7 |

**Table 4.** Descriptive Analysis of Ease of Use

| | Question 7 | Question 8 | Question 9 | Question 10 | Question 11 | Question 12 |
|---|---|---|---|---|---|---|
| **Mean** | 5.15 | 4.57 | 5.85 | 5.63 | 5.2 | 5.89 |
| **Standard Error** | 0.43 | 0.53 | 0.34 | 0.32 | 0.40 | 0.43 |
| **Median** | 5 | 6 | 6 | 6 | 6 | 7 |
| **Mode** | 7 | 7 | 7 | 7 | 6 | 7 |

From the result we can concluded that the average score from the prototype is around 4.5 to 6.3. This is because the users are unfamiliar with the game environment and new to the system. The median of every question is around five to seven. From the result we can conclude that this game will help paced the history revision and the system was easy to use in the process. We also include a few questions inside the questionaires to be fill up by the users. The questions are about the most negative and positive aspect of the game.

From the feedback, we found that the most negative aspect is the violent and the use of gun in the game and it's not really appreciated by the users. Meanwhile the most positive aspect is about the effectiveness of the game to help the users to revise the history subject. They can revise the subject quickly and much more fun compare to the conventional methods. With this feedback, we are confident that this game can make a contribution to help the student to do revision for history subject in school.

## 4 Conclusion

This prototype has been developed to achieve four main objectives. The first objective is to identify the requirements for an educational game. This objective has been attained by using several methods and techniques such as literature review, e-brainstorming, observation / interview and the requirements have been derived. The next objective of this project is to develop an educational game prototype to help the student to revise history subject. Based on the requirement identified earlier, the prototype has been developed using the Game Maker software and all the processed involved were discussed in this paper.

The third objective is to test the functionality of the educational game prototype. To accomplished this, we used the black box testing method to test the functionality of the educational game. The testing showed us that the educational game prototype is functional and ready to be used. The last objective of the study is to measure the usefulness and ease of use of the prototype. Using PUEU test, the result shows that this educational game has fulfilled the user requirement, easy to be use by the user and it is usefull to the student in learning.

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